# NISHAL STANISLAUS SILVA, *PhD*

Machine Learning Research Engineer | Deep Learning, Time-Series, Pattern Detection | Applied ML

*Canadian Permanent Resident, Eligible to work in Canada without sponsorship*

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7th October 2026

**Re: Senior Machine Learning Engineer (Fraud) - Affirm (Remote Canada)**

Dear Hiring Manager,

I am applying for the Senior Machine Learning Engineer (Fraud) role on the ML Fraud team at Affirm. Fraud models that decide on a live transaction have to be right and fast at the same time, and that pairing is the core of my record: I am an ML engineer with a PhD from the University of Trento (2025) and 10+ years across industry and academia, and my most recent deployed detector made real-time decisions in 14 ms on a Raspberry Pi 4 at F1 0.76, 74× faster than the DTW baseline it replaced (JAES 2026).

You need someone who can take a model from idea to production and keep it healthy. Here is how my work maps to that:

- **Risk models on structured data.** At Forestpin I designed and deployed ML and statistical pipelines for financial forensics, compliance monitoring and risk detection on large structured enterprise datasets, and built the SQL + Python scoring services that flagged anomalies for triage in production.
- **Low-latency decision systems.** On the EU-funded MUSMET project I owned the full pipeline, from feature engineering to training, deployment and monitoring, with inference held under 30 ms. At MAS Holdings I shipped live accept/reject inspection (OpenCV with PLC reject) that cut inspection time by 99.5%.
- **Offline experiments before production.** I design baselines and ablations to choose what ships, and recently benchmarked 6 scoring methods over 100,000 trials for a binary classifier (best ROC-AUC 0.822 vs. 0.599 baseline).
- **Engineering practice.** I introduced CI/CD and code review at MAS, mentored engineers, orchestrate pipelines in Airflow, and use Claude Code daily as part of my development workflow.

To be direct about the gaps: gradient-boosted tree libraries (LightGBM, XGBoost, CatBoost) and Spark are not yet in my daily toolkit, and I have not modelled transaction graphs. My structured-data scoring at Forestpin and my Python, scikit-learn and PyTorch work make these a fast ramp rather than a restart.

Affirm's FY2026 annual report describes fraud models that draw on roughly 350 data points to make a near-instantaneous decision on whether to block a transaction, at a scale of $50.2 billion in GMV. That is the problem I want to work on: feature-rich models under a hard latency budget, tuned as a balance between fraud loss, customer experience and conversion. In my first months I would aim to own one model end to end, from features and offline backtests to a monitored production rollout.

I am a Canadian permanent resident living in Ontario, eligible to work without sponsorship, available immediately, and glad to attend in-person onboarding. I would welcome the opportunity to discuss how my experience in real-time ML and risk scoring can support the ML Fraud team at Affirm.

Sincerely,

**Nishal Stanislaus Silva**, PhD